

Applied and Buddy Group Engagement Report

First Activity

- Week number: 2
- Applied class date and time: 04/08/2026 2:00pm – 4:00pm
- TA name: Gary Zheng

Key Learning 1

I learned how different sorting algorithms, including Bubble Sort, Insertion Sort, and Selection Sort, organise data using different approaches. Working through these algorithms with my Buddy Group helped me better understand how elements are compared, swapped, or shifted during the sorting process.

Key Learning 2

I learned how sorting algorithms can be applied to a practical problem through the Sort and Store case study. Our group designed a solution that temporarily stored incoming users and sorted them by timestamp before sending them to the database. We used an insertion-sort-style approach to order the timestamps, which helped me understand how a sorting algorithm can be adapted and used as part of a larger system rather than only for sorting a simple array.

Second Activity

- Week number: 9
- Applied class date and time: 29/09/2026 2:00pm – 4:00pm
- TA name: Gary Zheng

Key Learning 1

I learned how a Max Heap can be used to efficiently manage tasks based on priority. In the activity, we used heap insertion for adding a single task in $O(\log n)$, and bottom-up heapify for adding a batch of tasks in $O(n+m)$. This helped me understand why repeatedly inserting every task in a batch would be less efficient, and how choosing the appropriate heap operation can satisfy different time-complexity requirements.

Key Learning 2

I learned that a heap is efficient for accessing the highest-priority task, but becomes more challenging when an additional constraint is introduced. In the Task Manager, the highest-priority task may not fit within `minutes_left`, so we recursively searched the heap for the best task that could be completed in the available time. After finding it, we removed it and restored the heap property using `sink` or `bubble_up`. This helped me understand both the advantages and limitations of heaps when solving problems with multiple constraints.